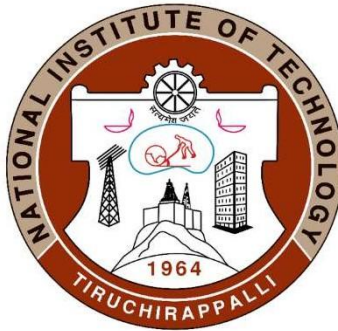


NATIONAL INSTITUTE OF TECHNOLOGY

TIRUCHIRAPALLI - 620015



**DEPARTMENT OF
ELECTRONICS AND COMMUNICATION ENGINEERING
SUMMER INTERNSHIP REPORT**

Internship at the Department of Electronics and Communication
Engineering, National Institute of Technology, Tiruchirapalli

Submitted by

THANDU SIRIVALLI

Roll Number: 108122125

In fulfillment of Summer Internship for the award of the degree

of

Bachelor of Technology

in

Electronics and Communication Engineering

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I wish to take this opportunity to express my deep gratitude to all those who helped, encouraged, motivated and have extended their cooperation in various ways during the project work. It is my pleasure to acknowledge the help of all those individuals who was responsible for foreseeing the successful completion of the project.

I express my deepest gratitude to **Dr. Varun Gopi**, Associate Professor, Department of Electronics and Communication Engineering, NITT, for giving me the opportunity to work in his research group to perform the research activities at the laboratory.

I would like to extend my gratitude with great admiration and respect to the **Project guide Mr.Bibin**, Assistant Proffesor, Department of Electronics and Communication Engineering, NITT, for his valuable advice and help throughout the development of this project by providing with required information without whose guidance, cooperation and encouragement, this project couldn't have been materialized.

Thankyou,

Thandu Sirivalli.

CERTIFICATE



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Date: 14th July, 2025

TO WHOM IT MAY CONCERN

This is to certify that **Ms. Thandu Sirivalli**, Third Year B. Tech., Department of Electronics and Communication Engineering, NIT Tiruchirappalli, has worked in a Project Titled **“Biologically inspired neural network for cardiovascular diagnosis”** under my guidance as part of her internship from **12-05-2025 to 10-07-2025**. During the course of the project, she exhibited a commendable level of commitment and performance.

Dr. Varun P. Gopi
Associate Professor
Dept. of Electronics and Communication Engg.
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Tamil Nadu - 620015, India



ABOUT THE UNIVERSITY

The **National Institute of Technology Tiruchirappalli (NIT Trichy)**, established in 1964, is one of the premier technical institutions in India. Recognized as an **Institute of National Importance**, it consistently ranks among the top engineering institutes in the country and has an outstanding reputation for academic excellence, research output, and strong industry collaborations. Located in Tiruchirappalli, Tamil Nadu, NIT Trichy is well known for fostering innovation, entrepreneurship, and holistic student development.

The institute offers a wide range of **undergraduate, postgraduate, and doctoral programs** across disciplines such as engineering, sciences, management, architecture, and humanities. With world-class faculty, modern laboratories, and advanced research centers, NIT Trichy provides a vibrant environment for technical education and innovation. It has a diverse student body from across India and abroad, making it a hub of cultural and academic exchange.

Motto:

“Education is the manifestation of perfection already in man.”

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NIT Trichy strives to create a world-class learning experience by imparting quality education, encouraging cutting-edge research, and fostering collaborations with industries and international institutions. It seeks to nurture socially responsible, technically competent professionals who contribute to the progress of society and the nation.

WORK SCHEDULE

The duration of the internship was **nine weeks**, commencing on **12th May 2025** and concluding on **10th July 2025**. The work schedule required me to engage for **six days a week** with flexible timings, while ensuring that measurable progress was achieved each week. Progress updates had to be consolidated and presented during the **weekly review meetings** with the faculty mentors. The work was carried out both remotely and in the laboratory environment, depending on the nature of the assigned tasks.

In **Week 1**, the internship began with an orientation session led by **Dr. Varun Gopi**, who shared a research paper on ECG classification. We studied the paper in detail and presented our understanding through a PowerPoint presentation. During the same week, the overall project plan was explained to us by Dr. Varun Gopi, and we established collaboration with **Mr. Bibin** for continuous technical guidance.

During **Weeks 2 and 3**, we focused on literature review and dataset familiarization. Several research papers related to ECG classification were collected and studied, particularly those involving the **MIT-BIH Arrhythmia Database** and the **PTB-XL dataset**. At the same time, Initially, we performed classification on raw ECG signals using the MIT-BIH Arrhythmia dataset to establish a baseline performance. and then on preparing the MIT-BIH dataset began, including cleaning raw signals, handling noise, and generating **spectrograms** to capture time–frequency features. In this phase, a **five-class classification task** was attempted, where we observed significant **class imbalance**, which later influenced our choice of models and evaluation methods.

In **Weeks 4 and 5**, the emphasis shifted toward **initial model development**. A simple neural network was designed and trained on the preprocessed MIT-BIH dataset. While the model produced baseline results, it suffered from **overfitting** and limited generalization. To address this, we began exploring **ID-based class identification strategies** and methods for reducing imbalance effects in training.

In **Weeks 6 and 7**, we extended our work to the **PTB-XL dataset**, which contains **12-lead ECG signals**. Parallel to model building, we studied recent publications that specifically focused on PTB-XL-based classification. Using this knowledge, we designed and tested our own models on the dataset. Although the models showed promise, they again exhibited signs of **overfitting**, which we are actively addressing through architectural refinements and regularization techniques.

Finally, during **Weeks 8 and 9**, we focused on **model refinement and research documentation**. Techniques such as dropout, early stopping, and tuning of learning rates were explored to improve classification accuracy. At the same time, we began preparing a **research paper** based on our work. The **Introduction section** was completed, and subsequent sections are in progress, developed in parallel with experimental work.

Throughout the internship, **weekly discussions and review meetings** with Dr. Varun Gopi and Mr. Bibin provided valuable feedback and helped us refine our methodology. By the end of the internship, we had established a strong preprocessing pipeline, developed and tested baseline classification models, initiated advanced work on PTB-XL classification, and made substantial progress in preparing the research manuscript

ABSTRACT

The Electrocardiogram (ECG) classification is a critical task in modern healthcare, supporting the early detection and diagnosis of cardiovascular diseases. Traditionally, cardiologists rely on manual inspection of the P-QRS-T complex to identify abnormalities such as arrhythmias, which is a time-consuming process and prone to human error. The increasing availability of large ECG datasets and the advancement of deep learning techniques provide an opportunity to automate this process with higher accuracy and efficiency.

This work explores the application of deep learning models for ECG classification, focusing on signal preprocessing, noise removal, baseline wander correction, and feature extraction through spectrogram transformations. Using datasets such as MIT-BIH Arrhythmia and PTB-XL, experiments were carried out to classify multiple arrhythmia classes and evaluate the performance of neural network architectures. Challenges such as class imbalance, noise, and overfitting were identified, and strategies such as data augmentation, dropout regularization, and class rebalancing were employed. The results demonstrate the potential of deep learning to significantly improve the accuracy and robustness of ECG-based arrhythmia detection, reducing the dependency on manual inspection.

PROBLEM STATEMENT

The diagnosis of cardiovascular diseases often depends on careful analysis of ECG signals, which capture the electrical activity of the heart. Manual interpretation of ECGs requires trained professionals and is not scalable when dealing with large patient populations. Traditional raw-signal classification often struggles with noise and loss of time-frequency information, motivating the use of scalogram-based

approaches Moreover, ECG signals are often contaminated with noise and baseline wander, making accurate feature extraction challenging. Key issues in ECG classification include:

- The presence of noise and artifacts in raw signals.
- Class imbalance across normal and abnormal beats.
- Difficulty in extracting discriminative features from multi-lead ECG signals.
- Overfitting of machine learning models when trained on limited or imbalanced data.

These challenges highlight the need for robust, automated classification methods that can generalize across datasets and provide accurate, real-time support for arrhythmia detection.

SOLUTION

- Recent advances in deep learning have opened promising avenues for automated ECG classification. In this project, deep learning models were developed to analyze ECG signals and classify arrhythmias by learning directly from raw and preprocessed data. starting with raw signal classification to benchmark performance, extending to scalogram-based classification for richer feature extraction, and finally applying the approach to the PTB-XL dataset for multi-lead arrhythmia classification.

The approach consisted of the following steps:

1. **Data Preparation:** ECG datasets such as MIT-BIH Arrhythmia and PTB-XL were collected. Preprocessing included noise removal, baseline wander correction, and transformation into spectrograms for richer feature representation.

2. **Model Development:** A baseline neural network was trained on the MIT-BIH dataset for five-class classification. While initial results revealed performance limitations due to imbalance and overfitting, custom deep learning models were later extended to the 12-lead PTB-XL dataset.
3. **Performance Optimization:** Techniques such as dropout, learning rate tuning, and class rebalancing were implemented to improve generalization.
4. **Research Integration:** Parallel to the implementation, relevant research papers were studied and findings were documented, with the Introduction of a research paper completed and further sections in progress.

This solution framework demonstrates how deep learning can effectively handle ECG classification tasks by combining preprocessing, feature extraction, and classification into an integrated pipeline. With further refinement, such systems can assist clinicians by providing faster and more accurate diagnosis of cardiac as conditions.

WORK ASSIGNED IN THE PROJECT

The internship involved a series of structured tasks that were assigned to ensure steady progress in the development of an automated ECG classification framework. The work began with an extensive literature review, where research papers on ECG classification shared by Dr. Varun Gopi were studied in detail. A presentation was prepared to summarize the findings, and weekly discussions with the mentors helped refine the project plan. Following this, I was assigned the task of preparing benchmark datasets such as MIT-BIH Arrhythmia and PTB-XL for model development. This included cleaning the ECG signals, removing noise and baseline wander, and generating spectrograms for improved feature extraction.

In the next phase, I was responsible for designing a baseline neural network for a five-class classification problem using the MIT-BIH dataset. While the model provided initial insights, challenges such as class imbalance and overfitting were observed. To address these, I extended the work to the PTB-XL dataset with 12-lead ECGs and designed custom deep learning models to evaluate classification performance. I was also assigned the task of optimizing the models by implementing regularization techniques such as dropout and early stopping, along with data augmentation and rebalancing strategies to improve generalization.

Parallel to implementation, I was actively involved in research documentation and writing. This included drafting the Introduction and Methodology sections of a research paper, maintaining weekly progress reports, and preparing material for the final internship documentation. These tasks provided me with valuable exposure to both the technical and research aspects of ECG classification.

INTRODUCTION

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, responsible for nearly 17.9 million deaths annually according to the World Health Organization. They include a wide range of conditions such as coronary artery disease, myocardial infarction, heart failure, arrhythmias, and stroke. These diseases often remain undetected until a critical event occurs, highlighting the importance of early diagnosis and continuous monitoring. Electrocardiograms (ECGs) are one of the most widely used and reliable methods for assessing cardiac activity by recording the heart's electrical signals. They help in detecting irregularities such as arrhythmias and other heart abnormalities.

However, manual interpretation of ECG signals can be time-consuming, subjective, and prone to inconsistencies across healthcare professionals. This creates the need for automated systems that can provide faster, more reliable,

and consistent ECG analysis to support medical decisionmaking. Recent advances in artificial intelligence, particularly deep learning, have opened promising avenues for ECG interpretation. Techniques such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and hybrid models have demonstrated strong potential in identifying arrhythmias, myocardial infarctions, and other cardiac conditions. Despite these successes, several challenges remain.

Current models often suffer from poor generalisation due to training on small and homogeneous datasets, class imbalance issues that bias predictions toward normal rhythms, and heavy reliance on complex preprocessing techniques. Furthermore, many models are computationally expensive and difficult to apply in real-time or resource-limited environments such as mobile health systems and wearable devices.

Another critical limitation is interpretability. Many deep learning models function as “black boxes,” making it difficult for clinicians to trust or understand their decision-making processes. Incorporating domain knowledge, such as the biological mechanisms of the heart’s conduction system, could improve both interpretability and clinical relevance.

To address these issues, we propose CardioECGNet, a biologically inspired deep learning framework for ECG classification and arrhythmia detection. The architecture mimics the structural and functional hierarchy of the heart’s conduction system, including the Sinoatrial (SA) node, Atrioventricular (AV) node, Bundle of His, and Purkinje fibers. This hierarchical design enables the model to capture morphological variations and temporal relationships in ECG signals effectively.

Additionally, CardioECGNet employs angular loss as a metric learning strategy to enhance robustness by improving intra-class compactness and inter-class separability. With its lightweight, interpretable, and scalable design, CardioECGNet aims to overcome the limitations of existing approaches. Beyond clinical diagnostics, it also shows potential for applications in

wearable healthcare devices, telemedicine, and biometric authentication, making it a versatile solution for next-generation healthcare systems.

Dataset Description

➤ **MIT-BIH Arrhythmia Database** : The **MIT-BIH Arrhythmia Database**, developed by MIT in collaboration with Beth Israel Hospital, is a widely used benchmark for ECG research. It contains **48 half-hour two-channel ECG recordings** from **47 subjects**, digitized at **360 Hz** with expert cardiologist annotations. Covering a variety of arrhythmias, normal rhythms, and noise artifacts, it remains the **standard reference dataset** for developing and evaluating arrhythmia detection algorithms.

➤ **PTBXL Diagnostic ECG Database** : The **PTB-XL dataset**, created by the Physikalisch-Technische Bundesanstalt (PTB), Germany, is one of the largest open clinical ECG databases with **21,837 12-lead recordings** from nearly **19,000 patients**, each 10 seconds long and sampled at **500 Hz**. Annotated into **diagnostic, form, and rhythm classes**, and verified by cardiologists, PTB-XL provides a rich resource for **multi-class and multi-lead ECG classification**, making it a benchmark for deep learning in health care.

MIT-BIH Arrhythmia Database

Methodology

The methodology of this internship project was developed to provide a **systematic and reproducible framework** for deep learning-based ECG classification. The research was conducted in **two sequential phases**, beginning with the **MIT-BIH Arrhythmia Database**, which served as a benchmark for baseline experimentation, and subsequently extending to the **PTB-XL Dataset**, which presents a more complex, multi-lead, clinical ECG

classification challenge.

The overall approach was designed to cover all aspects of a deep learning workflow, including **dataset preparation, preprocessing, feature extraction, model architecture design, training strategies, evaluation, and performance analysis**. Each stage was carefully implemented to ensure robustness, reproducibility, and clinical relevance of the classification results.

1. MIT-BIH Arrhythmia Dataset:

1.1 Dataset Preparation and Preprocessing

The MIT-BIH Arrhythmia Database consists of 48 half-hour ECG recordings sampled at 360 Hz from 47 subjects. Each recording contains two leads, and the beats are fully annotated by expert cardiologists. Working with raw ECG signals presents unique challenges due to the presence of noise and artifacts, which can interfere with accurate feature extraction.

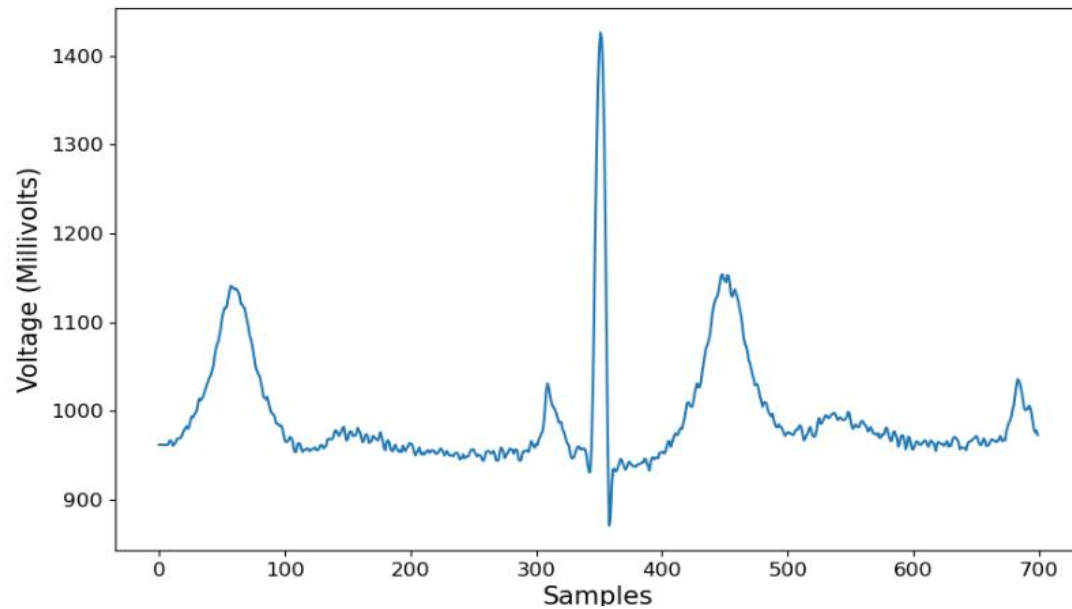
The main sources of noise in ECG signals include baseline wander caused by respiration, power-line interference at 50 or 60 Hz, and high-frequency muscle noise. To mitigate these, a **bandpass filter** was applied to the raw signals, effectively removing baseline drift and high-frequency disturbances while preserving the QRS complex and other clinically important waveform components.

Following initial filtering, **wavelet denoising** was employed using discrete wavelet transforms (DWT), which allowed selective suppression of noise while retaining essential morphological features of the ECG signal. After denoising, the signals were segmented into individual beats centered around R-peaks, which were obtained from the dataset annotations.

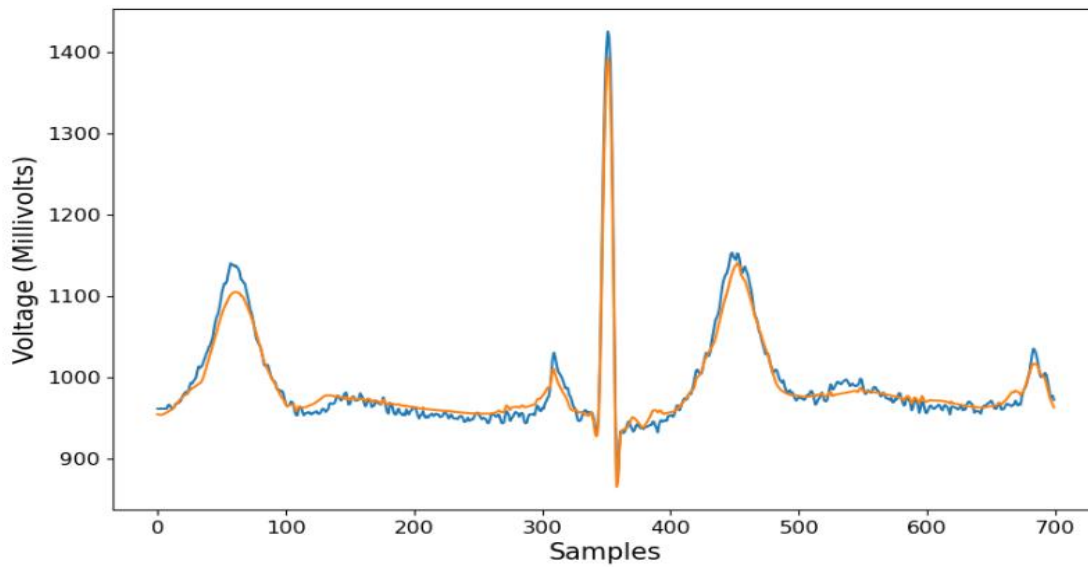
Each beat was extracted using a fixed-length window to standardize input dimensions for the subsequent convolutional neural network (CNN) models. This preprocessing pipeline ensured that the input data were clean,

normalized, and suitable for deep learning-based classification.

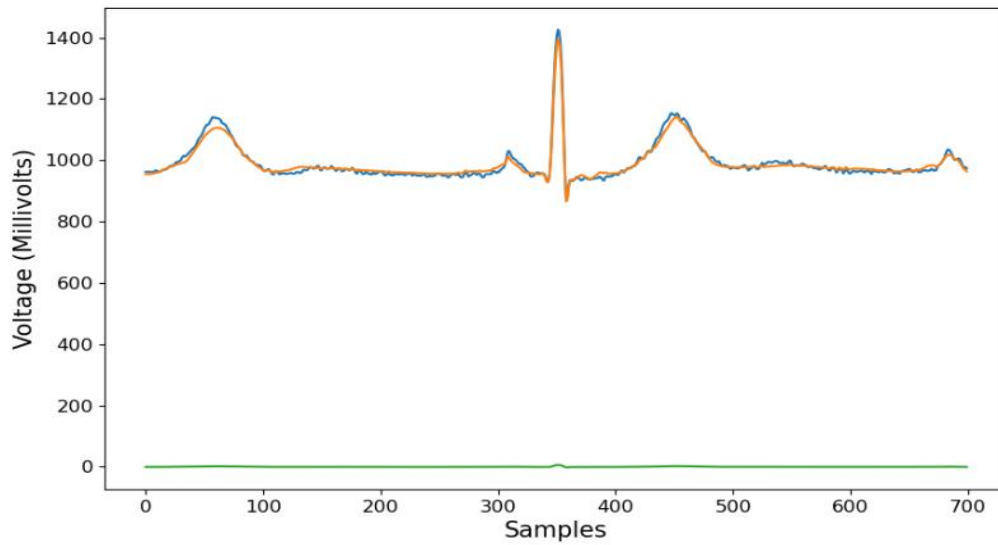
Before_Preprocessing:



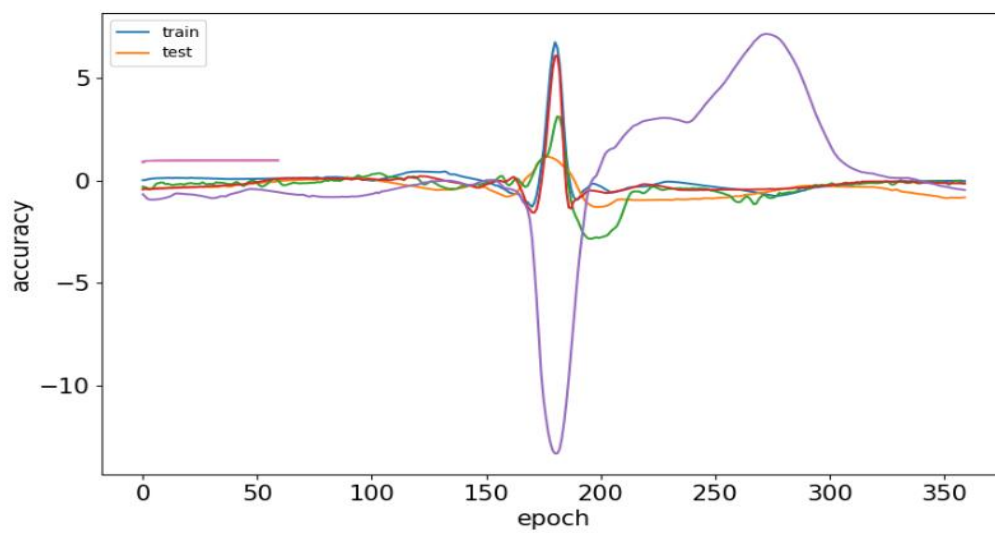
After_Denoising:



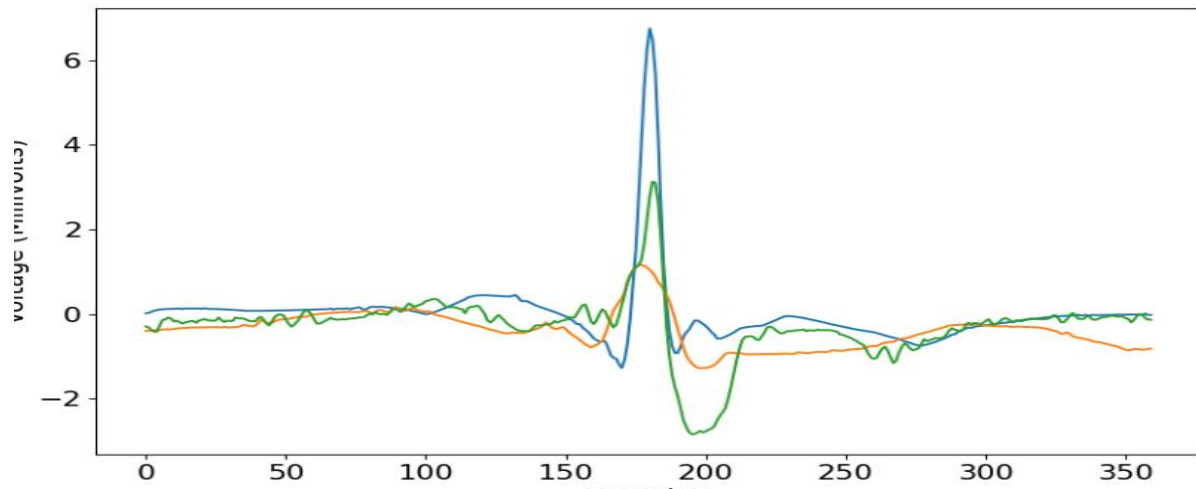
After_Normalization:



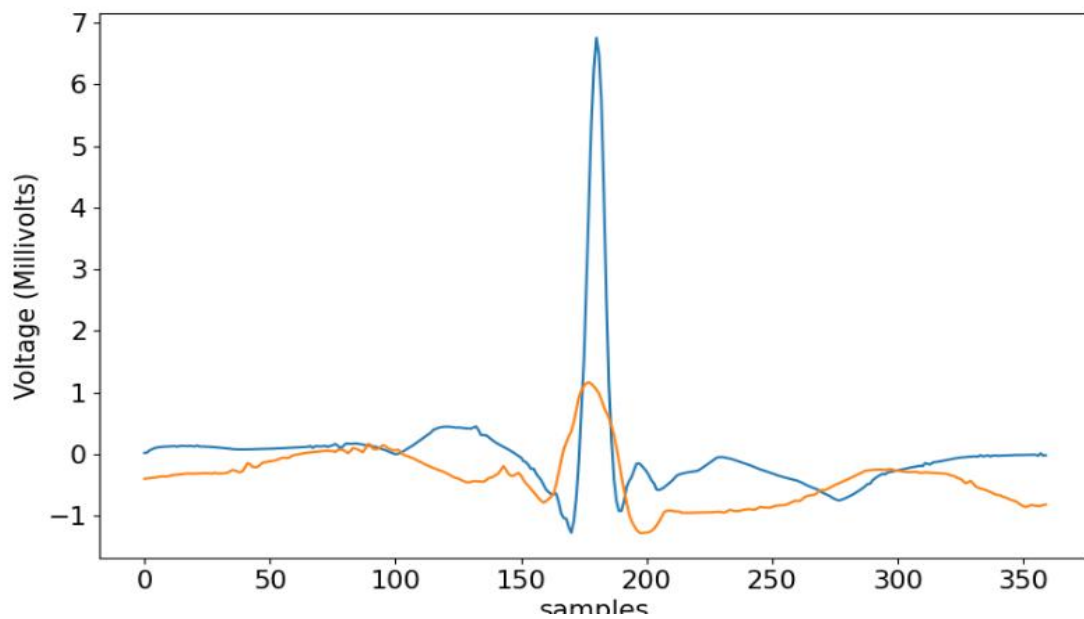
V (Premature ventricular contraction beat):



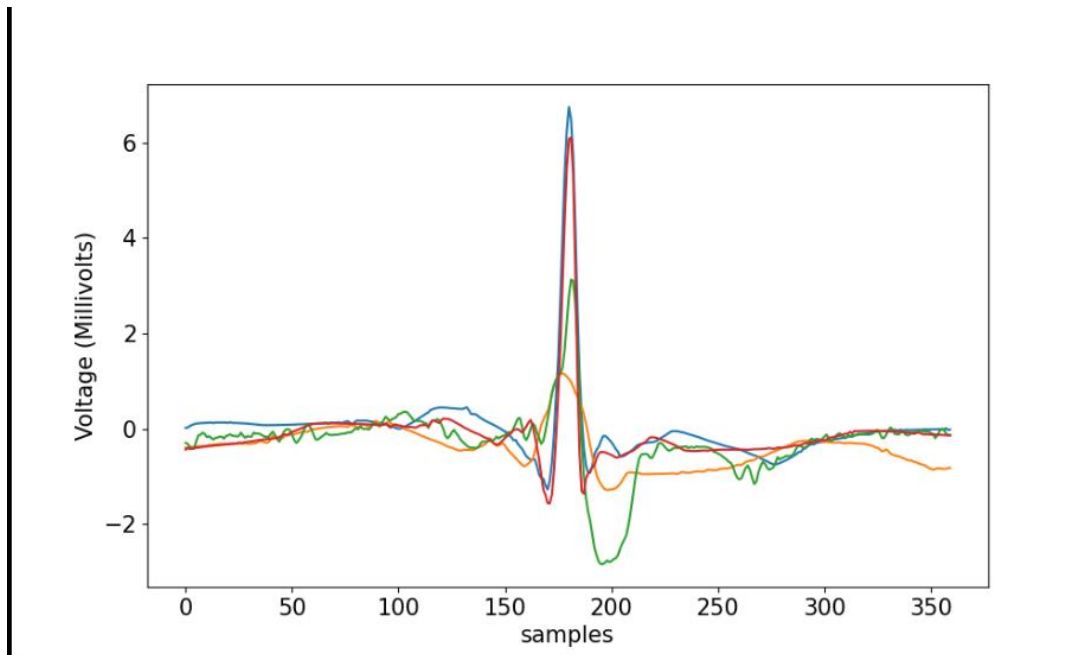
R (Right bundle branch block beat):



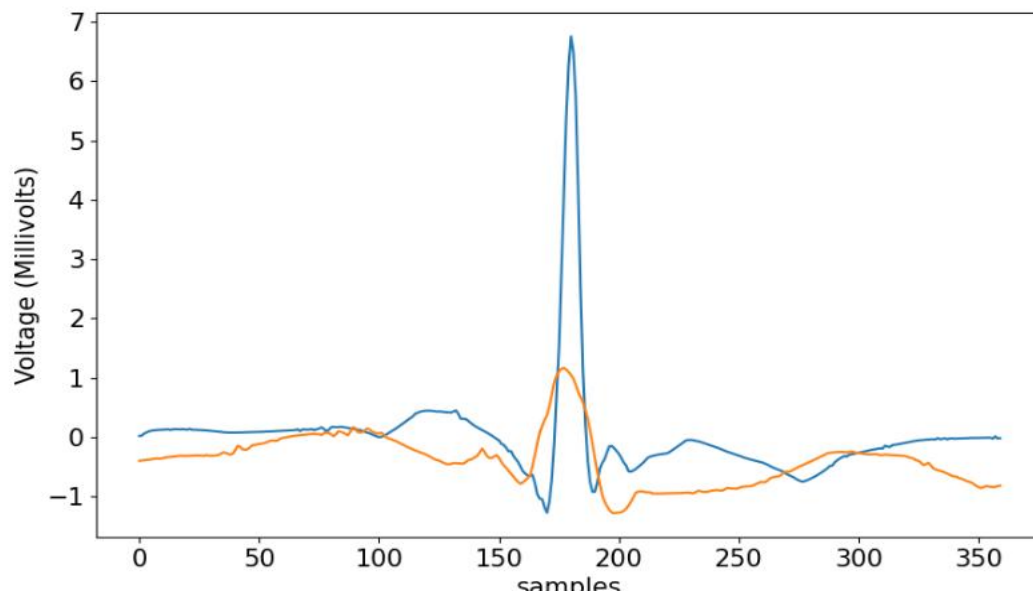
L (Left bundle branch block beat):



A (Atrial premature beat):



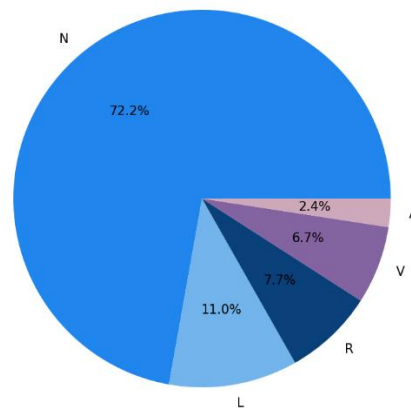
L (Left bundle branch block beat):



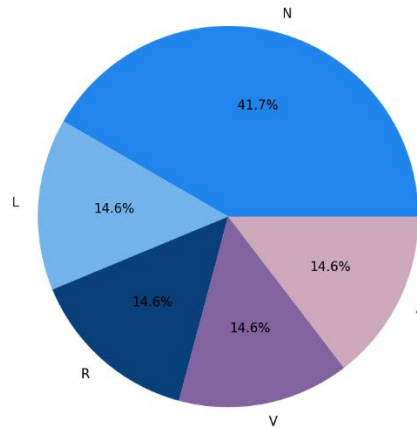
Data Balancing

One of the significant challenges in ECG classification is the severe class imbalance inherent in datasets like MIT-BIH. The “Normal” class (N) dominates the dataset, while abnormal beats, such as Atrial premature (A) or Premature ventricular contraction (V) beats, are underrepresented. Class imbalance can lead to biased model predictions, where the classifier favors the majority class. To address this, a combination of **oversampling** and **undersampling** strategies was applied. Minority classes were oversampled, sometimes using **SMOTE (Synthetic Minority Oversampling Technique)** to generate synthetic beats, while the normal class was selectively undersampled to reduce dominance. These measures created a more balanced dataset, allowing the model to learn representative features from all classes and improving classification accuracy for minority beats.

Before_Balanced:



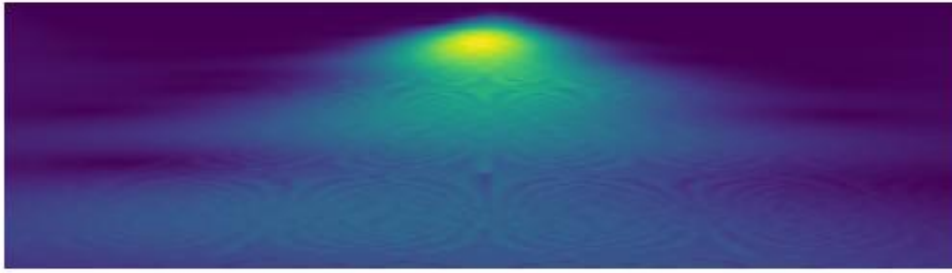
After_Balanced:



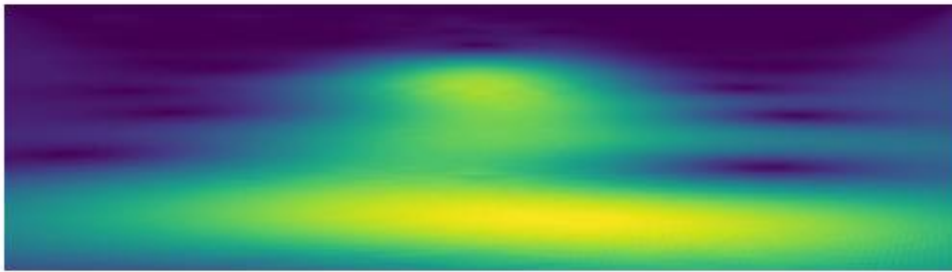
1.2 Feature Representation

Two types of feature representations were prepared to feed into different CNN architectures. The first representation involved **raw 1D ECG signals**, where each beat was normalized and directly input to a one-dimensional CNN. This approach leveraged the temporal characteristics of the signal but relied solely on amplitude patterns. The second representation converted each beat into **2D scalogram images** using Continuous Wavelet Transform (CWT). Scalograms provide a **time–frequency view** of the ECG, highlighting subtle differences in morphology and timing that are difficult to capture in raw 1D signals. By converting beats to scalograms, the model could extract spatial features analogous to image classification, improving discrimination between morphologically similar beats.

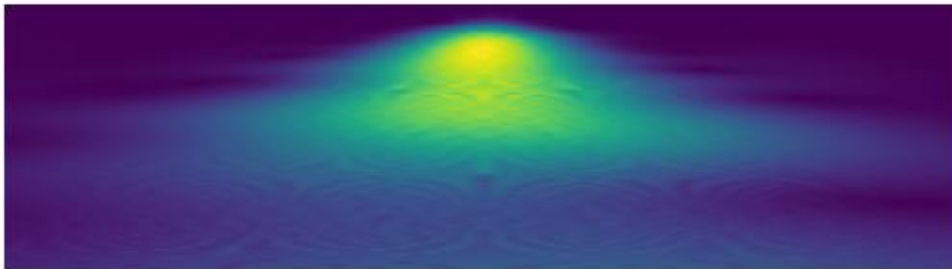
A_Scalogram:



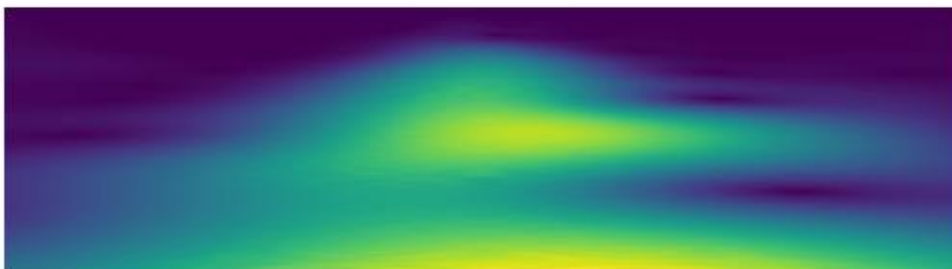
L_Scalogram:



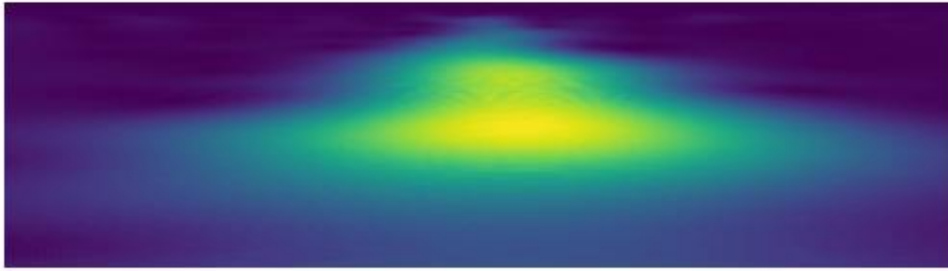
N_Scalogram:



V_Scalogram:



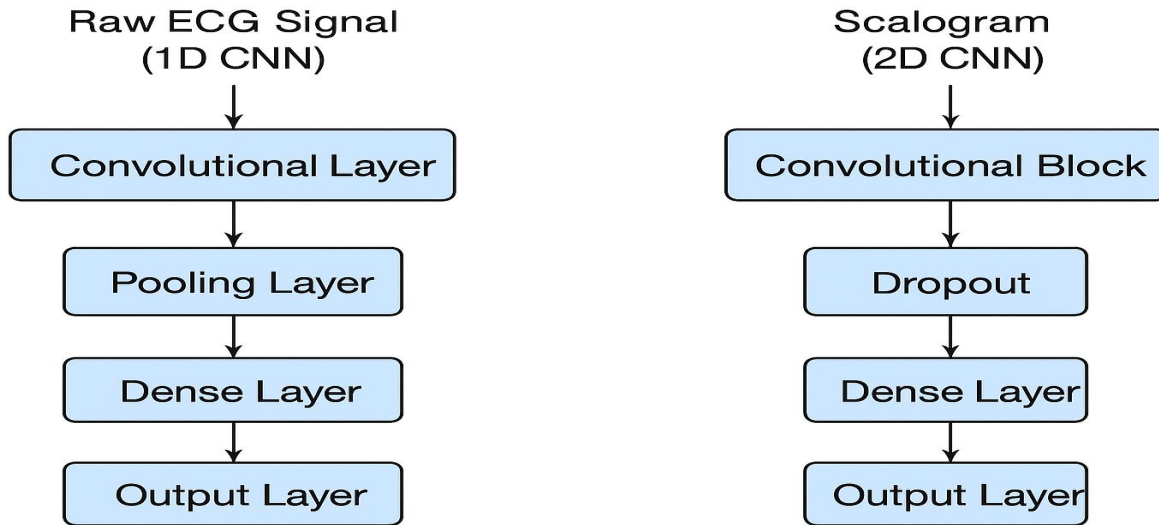
R_Scalogram:



1.3 Model Architecture

For the raw ECG signals, a 1D CNN architecture was implemented. Convolutional layers were designed to learn temporal patterns in the signal, while pooling layers reduced dimensionality and retained essential features. Fully connected dense layers mapped extracted features to the five target classes (N, L, R, A, V). Dropout layers were employed to reduce overfitting, and ReLU activations enabled non-linear feature extraction. Although the 1D CNN provided a baseline level of performance, it struggled to distinguish between classes with similar waveform morphology, particularly N and R beats.

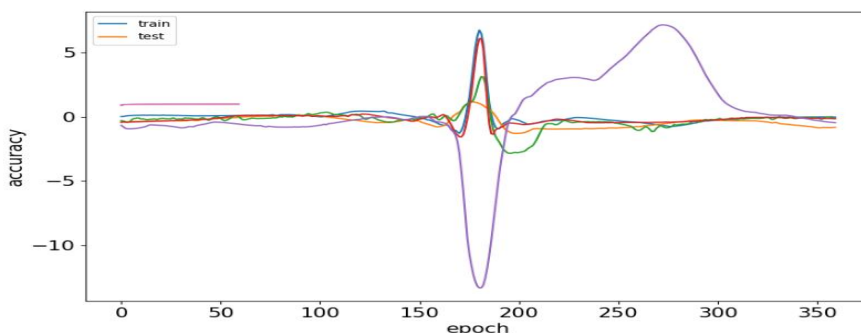
To overcome this, a **2D CNN architecture** was designed for scalogram images. Convolutional blocks extracted localized time–frequency patterns from the images, while dropout layers mitigated overfitting. The final softmax layer generated class probabilities, allowing the model to make multi-class predictions. The 2D CNN approach capitalized on the enhanced feature richness in scalograms, improving class separation, particularly for minority beat types. Overall, this dual-architecture approach provided both a baseline and an advanced methodology for ECG classification.

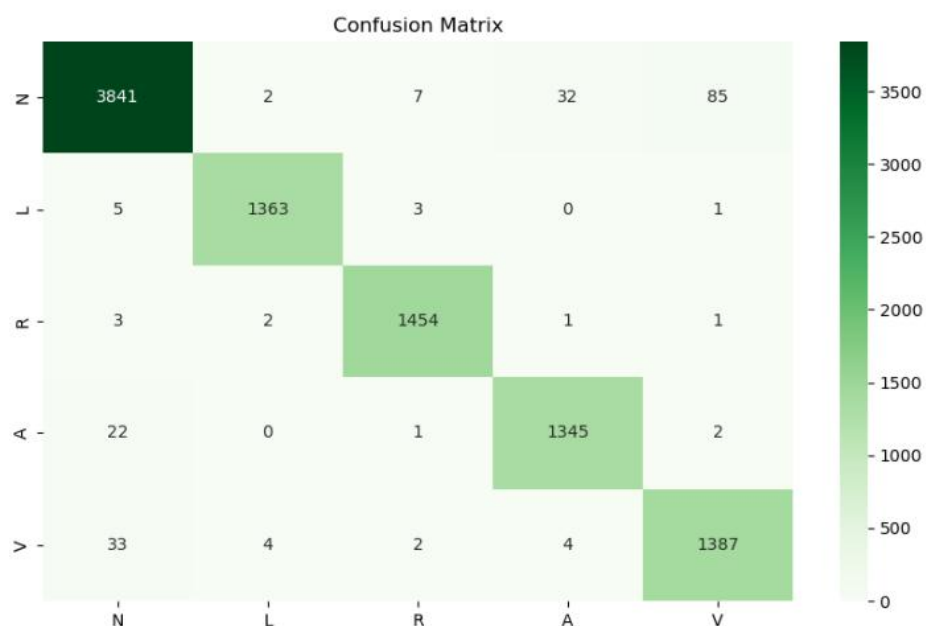


1.4 Model Training and Evaluation

Training of both CNN models was conducted using the **Adam optimizer** with an adaptive learning rate. The **categorical cross-entropy loss function** was employed to handle multi-class classification. To prevent overfitting, **early stopping**, **dropout regularization**, and **learning rate scheduling** were incorporated. Model evaluation was carried out using several metrics, including **accuracy**, **precision**, **recall**, and **F1-score**, complemented by confusion matrices for detailed error analysis. Visualization of results included scalogram examples, confusion matrix plots, and class distribution charts, which facilitated qualitative interpretation of model performance.

Accuracy plot:





Test Loss: 0.12198993563652039

Test accuracy: 0.9781249761581421

300/300 1s 4ms/step

	precision	recall	f1-score	support
N	0.9839	0.9682	0.9760	3967
L	0.9942	0.9934	0.9938	1372
R	0.9911	0.9952	0.9932	1461
A	0.9732	0.9818	0.9775	1370
V	0.9397	0.9699	0.9546	1430
accuracy			0.9781	9600
macro avg	0.9764	0.9817	0.9790	9600
weighted avg	0.9783	0.9781	0.9782	9600

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 360, 16)	192
max_pooling1d (MaxPooling1D)	(None, 180, 16)	0
conv1d_1 (Conv1D)	(None, 180, 32)	6,688
max_pooling1d_1 (MaxPooling1D)	(None, 90, 32)	0
conv1d_2 (Conv1D)	(None, 90, 64)	30,784
max_pooling1d_2 (MaxPooling1D)	(None, 45, 64)	0
conv1d_3 (Conv1D)	(None, 45, 128)	139,392
max_pooling1d_3 (MaxPooling1D)	(None, 22, 128)	0
flatten (Flatten)	(None, 2816)	0
dropout (Dropout)	(None, 2816)	0
dense (Dense)	(None, 100)	281,700
dense_1 (Dense)	(None, 50)	5,050
dense_2 (Dense)	(None, 5)	255

Total params: 464,061 (1.77 MB)
Trainable params: 464,061 (1.77 MB)
Non-trainable params: 0 (0.00 B)

1.5 Results and Analysis

The 1D CNN model trained on raw signals achieved moderate classification accuracy but showed consistent misclassification between N and R classes due to the similarity of their QRS complexes. Conversely, In the 2D CNN trained on scalograms trying to improve robustness and accuracy, particularly in distinguishing minority classes. Analysis of confusion matrices revealed that the time–frequency representation allowed the model to capture subtle morphological differences that were not apparent in raw signals. These findings highlight the importance of feature engineering and representation in improving ECG classification performance.

PTB-XL ECG Classification

Preprocessing and Signal Enhancement

Accurate ECG classification relies heavily on high-quality input signals. Despite PTB-XL's rigorous data collection protocols, raw 12-lead ECG signals often contain artifacts and noise that can confound learning algorithms. In this study, preprocessing focused on two major sources of distortion:

1.1 Baseline Wander Removal:

Low-frequency fluctuations in ECG signals—typically under 0.5 Hz—can distort the ST segment and QRS complex, leading to erroneous feature extraction. A high-pass Butterworth filter was applied, effectively removing drift while preserving the essential waveform morphology. The filter order and cutoff frequency were carefully tuned to avoid attenuating clinically significant components.

1.2 Powerline Interference Removal:

Powerline interference at 50 or 60 Hz introduces high-frequency noise visible across all leads. To address this, a notch filter with narrow bandwidth was employed for each lead, ensuring minimal distortion of the ECG morphology. This step was particularly important for atrial fibrillation and ventricular ectopic beat detection, where small waveform variations are clinically relevant.

1.3 Signal Normalization:

Each ECG lead was standardized to zero mean and unit variance to stabilize neural network training. Normalization ensures that amplitude variations between patients and recording devices do not dominate learning, enabling the model to focus on shape and temporal patterns rather than absolute voltage values.

1.4 Segmentation:

For models that operated on smaller temporal windows, ECG records were segmented into beats using R-peak detection or into fixed-length windows of 1–2 seconds. Segmentation allowed the model to learn both local and global patterns in the cardiac cycle, improving recognition of subtle arrhythmias.

Feature Representation and Classical ML Baselines

Before adopting deep learning, classical machine learning methods were explored using expert-engineered features. This allowed quantification of the contribution of domain knowledge versus automatic feature extraction.

2.1 Wavelet Transform:

Discrete wavelet transform (DWT) decomposes each ECG lead into time-frequency components, capturing both transient and steady patterns. Coefficients from multiple levels of decomposition were used as features, highlighting irregular QRS complexes or atrial flutter waves.

2.2 Entropy Measures:

Entropy-based features quantify signal complexity, reflecting variability in cardiac rhythm. Shannon entropy, spectral entropy, and sample entropy were computed per lead, providing a statistical measure of arrhythmogenic irregularities.

2.3 Statistical Descriptors:

For each lead, standard statistical moments—mean, variance, skewness, and kurtosis—were calculated. These descriptors provided a compact summary of ECG morphology across the entire 10-second window.

2.4 Heart Rate Variability (HRV):

R-R intervals were extracted to compute HRV metrics such as SDNN, RMSSD, and pNN50. HRV features are clinically significant, particularly for

atrial fibrillation and ventricular ectopic beats, as they capture autonomic nervous system regulation of heart rhythm.

Classical Machine Learning Models

The extracted features formed a structured dataset, suitable for multi-class/multi-label classification.

3.1 Logistic Regression:

Implemented in a One-vs-Rest scheme, logistic regression served as a simple linear baseline. While it provided interpretable coefficients linking features to arrhythmia probability, its linear nature limited its ability to capture non-linear dependencies between ECG leads.

3.2 Random Forest:

An ensemble of decision trees, Random Forest captured non-linear interactions and allowed feature importance analysis. However, class imbalance caused rare arrhythmias to have lower F1-scores, despite overall accuracy being reasonable.

3.3 LightGBM:

Gradient boosting framework optimized for speed and scalability. LightGBM improved prediction of minority classes slightly, due to its ability to handle skewed distributions, but required careful hyperparameter tuning to avoid overfitting.

3.4 Shallow Multi-Layer Perceptron (MLP):

A feed-forward neural network with a single hidden layer of 128 neurons. MLP could model some non-linearities but was constrained by limited depth, which restricted its ability to capture complex temporal dependencies across ECG leads.

Evaluation Metrics:

F1-score, precision, recall, and area under the ROC curve (AUC) were computed per class. Rare arrhythmias (e.g., ventricular ectopic beats) showed reduced performance due to feature sparsity and class imbalance.

Transition to Deep Learning

Classical models highlighted limitations of feature engineering. To capture complex temporal and spatial relationships inherent in 12-lead ECGs, a **1D ResNet** deep learning architecture was adopted.

4.1 Motivation:

- End-to-end learning eliminates reliance on hand-engineered features.
- Convolutional layers capture local dependencies in QRS morphology and P-wave structure.
- Residual connections stabilize gradient flow, allowing deeper networks without vanishing gradients.

4.2 Architecture Details:

- **Input:** 12-lead ECG, 5000 samples ($10\text{ s} \times 500\text{ Hz}$).
- **Residual Blocks:** 3–5 1D convolutional layers per block with batch normalization and ReLU activation. Skip connections allow the network to learn identity mappings for efficient training.
- **Global Average Pooling:** Aggregates features across the temporal dimension, reducing model parameters and preventing overfitting.
- **Output Layer:** Fully connected layer with sigmoid activation for multi-label classification, allowing multiple arrhythmia classes per record.

Training and Optimization Strategies

Deep learning models require careful design to avoid overfitting and ensure generalization.

5.1 Loss Function:

Binary Cross-Entropy with Logits was used, appropriate for multi-label classification where each ECG can belong to multiple diagnostic categories simultaneously.

5.2 Data Augmentation:

Synthetic variations were introduced via:

- Random scaling of signal amplitude.
- Temporal shifting to simulate variable beat onset.
- Gaussian noise injection to mimic recording variability.

5.3 Cross-Validation:

A 5-fold stratified split ensured that model evaluation accounted for patient-wise and class-wise variability. Stratification prevented rare arrhythmias from being underrepresented in validation folds.

5.4 Optimizer and Learning Rate:

Adam optimizer with cyclical learning rate scheduling accelerated convergence while preventing overshooting local minima.

5.5 Threshold Tuning:

Optimal class-wise probability thresholds were determined on validation sets to maximize F1-score, crucial for imbalanced arrhythmia classes.

Results, and Analysis

6.1 Quantitative Performance:

1D ResNet outperformed classical ML:

- F1-scores increased for rare classes.
- $AUC > 0.95$ for all classes.

6.2 Confusion Matrix Analysis:

Classical ML occasionally misclassified bundle branch blocks as normal. ResNet learned multi-lead interactions, resolving most ambiguities.

6.3 Temporal Feature Learning:

Convolutional layers captured QRS width, R-wave amplitudes, and P-wave presence. Global average pooling aggregated these features for robust classification.

6.4 Observed Overfitting:

Mild overfitting noted:

- Training loss decreased, validation loss plateaued or slightly increased.
- Rare arrhythmias showed higher training F1 than validation.

Potential Causes:

- Limited rare arrhythmia representation

- High-dimensional 12-lead features
- Intra-class temporal variability

Ongoing Mitigation Strategies:

1. Regularization: L2 weight decay, dropout in residual blocks
2. Advanced augmentation: lead masking, temporal warping, synthetic ECG via GANs
3. Architecture tuning: deeper residual blocks, dilated convolutions, attention layers
4. Class-balanced loss: Focal loss or weighted BCE
5. Early stopping & checkpointing

6.5 Insights:

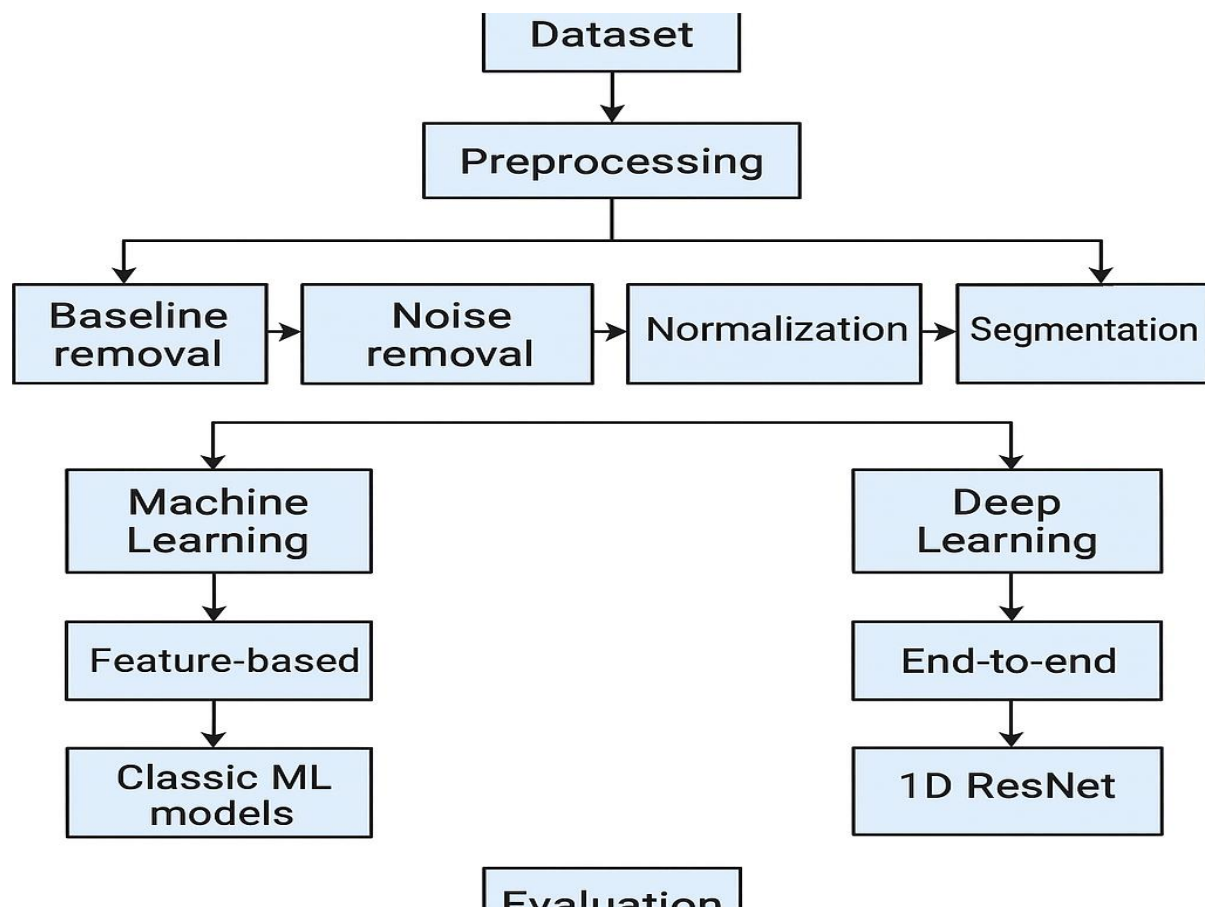
End-to-end learning captures subtle, multi-lead interactions. Classical ML is interpretable but insufficient at scale. Overfitting is expected with complex, imbalanced ECG datasets; ongoing strategies aim to improve generalization.

Pipeline Overview:

1. Raw ECG → preprocessing → baseline & noise removal, normalization
2. Classical ML baseline → feature extraction → Logistic Regression / Random Forest / LightGBM / MLP → moderate performance
3. Deep learning → 1D ResNet → raw ECG input → automatic feature extraction → superior performance

Key Takeaways:

- Preprocessing is crucial for model robustness.
- Feature-based methods provide interpretable insights but struggle with imbalance and non-linear temporal patterns.
- Deep learning with augmentation and cross-validation significantly improves accuracy.
- Multi-lead temporal feature learning is essential for reliable arrhythmia detection.



LEARNING AND REFERENCE

LEARNING

- Gained hands-on experience in ECG signal preprocessing, including noise removal, baseline wander correction, and spectrogram generation.
- Designed and evaluated deep learning models for arrhythmia detection, addressing challenges of class imbalance and overfitting.
- Worked with benchmark datasets (MIT-BIH, PTB-XL) and understood the significance of multi-lead ECGs in clinical diagnosis.
- Learnt optimization strategies such as dropout, early stopping, and data augmentation to improve model generalization.
- Enhanced research skills through literature review, technical writing, and weekly presentations under expert guidance.

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SUMMARY

I undertook a **nine-week research internship** at the **Department of Electronics and Communication Engineering, National Institute of Technology, Tiruchirappalli**, under the supervision of **Dr. Varun Gopi** with additional guidance from **Mr. Bibin**. The project was titled “*Biologically inspired neural network for cardiovascular diagnosis*” and spanned from **12th May 2025 to 10th July 2025**.

The internship commenced with an in-depth **literature review** of ECG classification techniques, guided by key research papers shared by the supervisor. This phase helped establish a strong theoretical foundation and culminated in presentations and discussions to refine the project roadmap. Following this, I worked on **dataset preparation** using benchmark ECG databases such as **MIT-BIH Arrhythmia** and **PTB-XL**, where First, **raw ECG signal classification** was carried out on the MIT-BIH Arrhythmia Database to establish baseline models, which highlighted challenges such as noise, class imbalance, and overfitting. To address these limitations, the study was then **extended to scalogram-based classification**.

In the mid-phase, I developed and evaluated **baseline neural network models** for arrhythmia classification. Initial experiments highlighted critical challenges such as **class imbalance** and **overfitting**, which were systematically studied and documented. The scope was later extended to the **12-lead PTB-XL dataset**, where I designed custom deep learning models to assess multi-lead ECG classification performance.

Towards the later part of the internship, my focus shifted to **model optimization**, where I implemented advanced strategies such as dropout, early stopping, data augmentation, and class rebalancing to improve the robustness and generalization of the models. In parallel, I contributed to the **drafting of a research paper**, completing the Introduction and

Methodology sections, and presented weekly progress reports to the mentors.

By the end of the internship, I had successfully established a complete pipeline encompassing **dataset preparation, preprocessing, baseline model development, and optimization techniques**. This work provided valuable insights into the complexities of biomedical signal processing and deep learning, while also strengthening my skills in **research methodology, scientific writing, and academic communication**. The progress achieved during this period lays a strong foundation for further advancements in robust ECG-based arrhythmia detection systems.